

Derivation of the UAT Lagrangian Parameters: From Fundamental Constants to ξ and λ

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1 Starting Point: The UAT Action

The UAT Lagrangian is

$$S_{\text{UAT}} = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{\lambda}{4} (\phi^2 - \eta^2)^2 - \frac{\xi}{2} R \phi^2 + \mathcal{L}_m \right].$$

Three unknown parameters appear: η , ξ , and λ .

2 Constraint 1: The Causal Limit

The Ivancho constant $\kappa_{\text{crit}} = 4.978$ sets the vacuum scale of the field:

$$\boxed{\eta = \kappa_{\text{crit}} = 4.978}.$$

3 Constraint 2: The Thermal Calibration Margin

The 7% noise margin determines the field value at recombination:

$$\phi_* = 0.07 \eta = 0.34846.$$

4 Constraint 3: The Quantum Braking Factor

The relation between k_{early} and ξ is

$$k_{\text{early}} = \frac{1}{1 - \xi \phi_*^2}.$$

With $k_{\text{early}} = 0.967$ and $\phi_* = 0.34846$,

$$\xi = \frac{1 - 1/k_{\text{early}}}{\phi_*^2} = \frac{-0.0341}{0.1214} \approx -0.2810.$$

5 Derivation of λ

The effective potential is $V_{\text{ef}}(\phi) = \frac{\lambda}{4}(\phi^2 - \eta^2)^2 + \frac{\xi}{2}R\phi^2$. The instability condition $V''_{\text{ef}}(\eta) = 0$ gives

$$2\lambda\eta^2 + \xi R_c = 0.$$

At recombination ($z \approx 1089$), the Ricci scalar is $R_c \approx 8\pi G \rho_{\text{rec}}$. The recombination density is

$$\rho_{\text{rec}} \approx (1.1 \text{ eV})^4 \approx 2.17 \times 10^{-111} M_{\text{Pl}}^4.$$

Solving for λ :

$$\lambda = -\frac{\xi R_c}{2\eta^2} = \frac{0.2810 \cdot 8\pi \cdot 2.17 \times 10^{-111}}{2 \cdot (4.978)^2} \approx 3.08 \times 10^{-112}.$$

6 Summary

$$\boxed{\eta = 4.978, \quad \xi = -0.2810, \quad \lambda = 3.08 \times 10^{-112}}.$$

These three numbers, together with the action S_{UAT} , constitute the complete mathematical foundation of the UAT framework.